

Project Concept Note

Project Title and Application Name PhishGuard AI

Problem Statement / Objective Phishing attacks remain one of the most widespread and damaging cybersecurity threats today. Everyday internet users and corporate employees frequently struggle to distinguish legitimate emails, SMS messages, or URLs from sophisticated malicious ones. This confusion often leads to credential theft, financial loss, and severe data breaches.

The objective of PhishGuard AI is to provide an intelligent, automated phishing detection tool that empowers users to verify suspicious content safely and effectively, while maintaining uninterrupted service through a robust fallback mechanism.

Target User and Use Case

- **Target Users:** Everyday internet users, corporate employees, and small business owners who regularly receive emails, SMS messages, and external links.
- **Use Case:** A user receives an urgent, suspicious email asking them to verify their account details. Instead of clicking the embedded link or replying, they copy the text or URL into PhishGuard AI. The application analyzes it in real-time, flags the threat level (e.g., High Risk), highlights specific suspicious elements, and offers actionable security recommendations on what to do next.

LLM Model and API Used

- **LLM Model:** Google Gemini AI (Large Language Model)
- **API Used:** Google Gemini API (integrated via the `google-generativeai` Python SDK, utilizing specialized system prompts for deep threat analysis).

Key Features of the Application

- **AI-Powered Analysis:** Leverages Google Gemini to deeply analyze the context, urgency, intent, and structure of provided text or URLs.
- **Rule-Based Fallback System:** Automatically switches to local rule-based heuristics (regex and keyword matching) if the AI service is temporarily unreachable, ensuring uninterrupted protection.
- **Risk Scoring & Confidence:** Provides a clear Risk Level (Low, Medium, High) alongside a quantitative Confidence Score (0-100%).
- **Detailed Explanations & Recommendations:** Offers comprehensive, human-readable reasoning for the prediction and specific security steps for the user to take.
- **Local History Management:** Saves all past analyses in a local SQLite database for easy retrieval, tracking, and auditing.
- **Downloadable PDF Reports:** Generates professional, offline PDF summaries of individual threat analyses using the ReportLab engine.
- **Responsive Web Interface:** Built with React, TypeScript, and Vite, providing a fast, intuitive experience across all devices.
- **Dockerized Architecture:** Fully containerized frontend and backend for seamless, platform-agnostic deployment.

Expected User Experience and Outcomes

- **User Experience:** Users interact with a clean, accessible web interface. Upon submitting a suspicious message, they experience a smooth, progressive reveal of the analysis results. The design clearly communicates the risk level using color-coded alerts and easy-to-read cards, building trust and understanding.
- **Outcomes:**
 - Increased user awareness and education regarding common phishing tactics.
 - Significant reduction in successful phishing attacks by intercepting threats before users interact with them.
 - Enhanced user confidence when dealing with unsolicited, urgent, or suspicious digital communications.